ModelIt!

Teacher's Guide

When Permafrost Melts, What Wakes Up?

Modeling Arctic Carbon Release and Climate Amplification

Quick Reference

Grade Level	Grade 11L2
NGSS Standard	HS-ESS2-4, HS-ESS2-6
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-ESS2-4, HS-ESS2-6

Use a model to describe how variations in the flow of energy into and out of Earth's systems result in changes in climate. Develop a quantitative model to describe the cycling of carbon among the hydrosphere, atmosphere, geosphere, and biosphere.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a quantitative model of the permafrost carbon feedback loop to predict how Arctic warming triggers cascading carbon release and climate amplification across interconnected Earth systems.
Disciplinary Core Idea	**ESS2.A Earth Materials and Systems / ESS2.D Weather and Climate** — Students model how energy flow variations (Arctic amplification) drive changes in the cryosphere that release stored carbon, connecting Earth material processes to climate system dynamics through the global carbon cycle.
Crosscutting Concept	**Cause and Effect / Stability and Change** — Students analyze how a small initial cause (Arctic warming) triggers a self-reinforcing feedback mechanism that can push the climate system past stability thresholds into a fundamentally different state.

CAST Testing Alignment

MULTIPLE CHOICE:

The Arctic has warmed approximately 3-4 degrees C above pre-industrial levels — 2-4 times faster than the global average. Permafrost thaw is releasing an estimated 1.5-2.5 billion tons of CO2 equivalent per year. Based on your model, which statement BEST describes the significance of this feedback?

CONSTRUCTED RESPONSE:

Using your model, explain why reducing all human fossil fuel emissions to zero might not be sufficient to stop permafrost carbon release if certain temperature thresholds are crossed. Reference the positive feedback loop, thermokarst formation, and the difference between human-controllable and natural carbon sources.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Beneath the frozen ground of the Arctic lies a climate time bomb: approximately 1,500 billion tons of organic carbon — dead plant material, animal remains, and ancient biomass — accumulated over thousands of years and preserved by permafrost. As the Arctic warms 2-4 times faster than the rest of the planet, this frozen carbon is beginning to thaw and decompose, releasing CO2 and methane into the atmosphere and creating what scientists fear could become an unstoppable positive feedback loop.

What Is Permafrost and What's Frozen In It?

Permafrost is ground (soil, rock, or sediment) that has remained at or below 0 degrees Celsius continuously for at least two years. It underlies approximately 25% of the Northern Hemisphere land surface, covering vast areas of Siberia, Canada, Alaska, and the Tibetan Plateau. Over tens of thousands of years, dead plant material that would normally decompose was instead frozen and preserved — much like a natural deep freeze. The total carbon stored is approximately 1,500 billion tons — roughly twice the amount currently in the atmosphere. Some of this carbon has been frozen for over 40,000 years, preserved since the last ice age.

The Thaw Process and Carbon Release

As Arctic temperatures rise, the active layer — the surface zone that thaws each summer — deepens progressively. Currently, the active layer is deepening by 2-3 cm per year in many regions. When permafrost thaws, ancient organic matter is exposed to microbial decomposition for the first time in millennia. In aerobic (dry) conditions, microbes release CO2. In anaerobic (waterlogged) conditions — particularly in thermokarst lakes — microbes produce methane, which is approximately 80 times more potent as a greenhouse gas than CO2 over a 20-year period. Current estimates suggest Arctic permafrost regions release 1.5-2.5 billion tons of CO2 equivalent annually.

The Positive Feedback Loop

The permafrost carbon feedback is a textbook example of a climate positive feedback loop. Step 1: Rising temperatures thaw permafrost. Step 2: Thawed organic carbon is decomposed by microbes, releasing CO2 and methane. Step 3: These greenhouse gases trap more heat in the atmosphere. Step 4: Additional warming causes more permafrost to thaw. This cycle is self-reinforcing — each round of thaw produces warming that drives the next round. The critical question is whether this feedback can become self-sustaining: could permafrost carbon release alone generate enough warming to continue thawing even if all human emissions stopped? Models suggest this threshold may be crossed at 4-6 degrees C of Arctic warming.

Thermokarst and Abrupt Thaw

Gradual surface thaw is only part of the story. Thermokarst — the collapse of ice-rich permafrost into sinkholes, lakes, and landslides — can expose deep permafrost to warm water and air almost instantaneously. Thermokarst lakes act as 'hot spots' of carbon release because warm water accelerates thaw far faster than warm air alone. In Siberia, massive craters tens of meters wide have appeared as underground methane pockets explode through thinning permafrost. These abrupt thaw events can mobilize carbon thousands of years old in days rather than decades. Recent research suggests abrupt thaw through thermokarst could double the amount of permafrost carbon released by 2100 compared to models that only account for gradual surface thaw.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify six components of the permafrost carbon feedback system: Arctic Temperature Anomaly, Permafrost Thaw Depth, Carbon Release Rate, Atmospheric Greenhouse Gas Level, Global Warming Rate, and Thermokarst Formation Rate — distinguishing the external temperature driver from internal feedback responses.

E **Establish Relationships**

Students determine that Arctic Temperature Anomaly drives Permafrost Thaw Depth, which determines Carbon Release Rate, which increases Atmospheric Greenhouse Gas Level, which amplifies Global Warming Rate, which feeds back to increase Arctic Temperature Anomaly — completing the positive feedback loop. Thermokarst Formation accelerates the entire cycle by exposing deep permafrost rapidly.

V **Visualize & Model**

Students build a computational model showing the positive feedback loop and how it amplifies initial warming through cascading carbon release, with visualization of nonlinear acceleration when thermokarst processes activate.

E **Evaluate Outcomes**

Students run moderate, current trajectory, and runaway scenarios to identify the Arctic temperature threshold at which the feedback loop becomes self-sustaining and to quantify the additional warming from permafrost carbon.

R **Revise & Extend**

Students add Abrupt Methane Release, Albedo Change, or Microbial Community Shift to explore additional feedback mechanisms and their interactions with the primary carbon release loop.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Buried beneath the Arctic is twice as much carbon as exists in the entire atmosphere. For thousands of years, it's been frozen. Now it's waking up."*

Watch For

Show satellite images of thermokarst lake expansion in Siberia over 20 years. Show a video of methane bubbling from a frozen lake being lit on fire. Ask: Where is that gas coming from?

Slide 2: Learning Objectives

3 min

Say: *"Today you're modeling a feedback loop that could determine whether we can actually stop climate change — even if we wanted to."*

Watch For

Have students read objectives. Pre-teach 'positive feedback loop' — emphasize that 'positive' means self-reinforcing, not good. Draw the loop on the board: warming → thaw → release → warming → ...

Slide 3: Big Question

3 min

Say: *"1,500 billion tons. That's twice what's already in the atmosphere. It's been frozen since the last ice age. What happens when we turn off Earth's freezer?"*

Watch For

Scale visualization: stack boxes representing carbon. One stack for the atmosphere. Make the permafrost stack twice as high. Let that sink in. Ask: What if even 10% of the big stack enters the atmosphere?

Slide 4: LEVER Framework

2 min

Say: *"We're modeling a feedback loop — a system that amplifies itself. The key question is: can it become unstoppable?"*

Watch For

Preview each LEVER step. Emphasize that this is fundamentally different from other climate models because the carbon source cannot be turned off by human decision — once permafrost thaws, the microbes decompose regardless of policy.

Slide 5: Activity 1: Components

8 min

Say: *"One external driver — temperature. Five internal responses. But notice: the last internal response feeds back to the first. That's what makes this a loop."*

Watch For

Guide sorting of external vs. internal components. Only Arctic Temperature Anomaly is external. Then challenge students: does Global Warming Rate feed back to affect Arctic Temperature? Yes — completing the loop.

Slide 6: Activity 2: Connections

8 min

Say: *"Trace the loop: warming melts permafrost, melting releases carbon, carbon traps heat, heat melts more permafrost. Where can you break this chain? Can you break it?"*

Watch For

Help students map every positive link in the loop. Key insight: every connection is positive, which means the loop is self-reinforcing. Discuss: at what point is breaking the loop impossible?

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test three futures. Moderate warming — manageable? Current path — dangerous? Runaway — unstoppable? Your model will tell us."*

Watch For

Students predict the self-sustaining threshold before running simulations. Compare carbon release rates across scenarios. Identify the temperature at which the feedback generates more warming than human emissions can offset.

Slide 8: Discoveries

5 min

Say: *"Your model just showed why scientists call permafrost the 'sleeping giant' of climate change. Once it fully wakes up, we can't put it back to sleep."*

Watch For

Lead discussion on why permafrost carbon is a fundamentally different climate threat than fossil fuels. Connect to the urgency of emission reductions before tipping points are crossed.

Slide 9: STEM Challenge

12 min

Say: *"We can't refreeze the Arctic. But we can watch it thaw — and sound the alarm if the feedback loop starts running away. Design the system that catches it."*

Watch For

Groups design monitoring networks combining ground sensors, satellite data, and atmospheric sampling. Present as proposals to an Arctic science consortium.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Arctic Temperature Anomaly
- **INTERNAL:** Permafrost Thaw Depth, Carbon Release Rate, Atmospheric Greenhouse Gas Level, Global Warming Rate, Thermokarst Formation Rate

Relationships

- **Arctic Temperature Anomaly to Permafrost Thaw Depth:** POSITIVE (+) — Higher Arctic temperatures directly increase the depth of the active layer and the rate of permafrost degradation. Each 1 degree C of warming deepens the active layer and extends the thaw season, exposing more ancient organic carbon to decomposition.
- **Permafrost Thaw Depth to Carbon Release Rate:** POSITIVE (+) — Deeper thaw exposes more organic material to microbial decomposition, directly increasing Carbon Release Rate. The relationship is nonlinear because deeper permafrost layers contain older, more concentrated carbon deposits — and thermokarst can expose them suddenly.
- **Carbon Release Rate to Atmospheric Greenhouse Gas Level:** POSITIVE (+) — Higher carbon release from permafrost adds CO2 and methane to the atmosphere beyond what human emissions contribute. This is an additional source that climate models must account for — and it cannot be regulated by emission reduction policies.
- **Atmospheric Greenhouse Gas Level to Global Warming Rate:** POSITIVE (+) — Higher greenhouse gas concentrations increase radiative forcing, trapping more thermal energy and accelerating Global Warming Rate. Each additional ton of CO2 or methane from permafrost contributes to warming just as much as a ton from a power plant.
- **Global Warming Rate to Arctic Temperature Anomaly:** POSITIVE (+) — Due to Arctic amplification (ice-albedo feedback, atmospheric circulation patterns), global warming is amplified 2-4 times in the Arctic. This closes the positive feedback loop: warming causes thaw, thaw causes more warming, amplified warming causes more thaw.

Key Discoveries

- Arctic permafrost contains approximately 1,500 billion tons of carbon — twice the amount currently in the entire atmosphere — frozen in organic matter accumulated over thousands to hundreds of thousands of years
- The permafrost carbon feedback creates a genuine positive feedback loop: thaw releases greenhouse gases that cause warming that causes more thaw — and unlike human emissions, this process cannot be turned off with policy decisions
- Thermokarst collapse dramatically accelerates thaw by exposing permafrost to warm lake water and air, creating localized carbon release hotspots that can emit methane explosively
- Most climate models have historically underestimated permafrost carbon feedback, meaning current emission reduction targets may need to be more aggressive to account for this additional warming source

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Permafrost is permanent"	Despite the name, permafrost is not permanent — it is defined only by temperature and time (below 0 degrees C for 2+ years). It formed during colder climate periods and is maintained only as long as temperatures remain low enough. As the Arctic warms, permafrost that has been frozen for thousands of years is thawing from the surface down, and this process is accelerating.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Permafrost thaw releases carbon slowly over thousands of years"	While gradual surface thaw does release carbon slowly, thermokarst processes can release ancient carbon rapidly — over years to decades rather than millennia. Thermokarst lake formation, coastal erosion of permafrost cliffs, and explosive methane release can mobilize deep carbon that would take centuries to reach through surface thaw alone. The model shows that abrupt thaw is the wild card that could dramatically accelerate the timeline.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"There's nothing we can do about permafrost thaw"	While we cannot directly refreeze the Arctic, we can reduce the severity of permafrost thaw by aggressively reducing greenhouse gas emissions to limit total warming. The difference between 1.5 degrees C and 4 degrees C of global warming determines whether the permafrost feedback remains manageable or becomes self-sustaining. Additionally, monitoring networks can provide early warning of acceleration, and some experimental approaches (like reflective ground covers and Pleistocene rewilding) are being tested to slow thaw locally.